

1. Before using the ROS feature pack, it is necessary to ensure that the firmware library has been installed. You can refer to the third section of the "Python Basic Control Course" on installing the Rosmaster driver library

2. Create, initialize, and compile a ROS workspace (using driver-ws as an example)

Input following command in the terminal

```
mkdir yahboomcar_ws
cd yahboomcar_ws
mkdir src
cd src
```

3. Copy and paste all the feature pack folders in the folder into the src folder of the ROS workspace, then return to driver-ws.

Input following command in the terminal

```
colcon build
```

```
sunrise@ubuntu:~$ cd yahboomcar_ws/
sunrise@ubuntu:~/yahboomcar_ws$ colcon build
Starting >>> openslam_gmapping
Starting >>> rf2o_laser_odometry
Starting >>> yahboomcar_base_node
Starting >>> yahboomcar_bringup
Finished <<< openslam_gmapping [3.73s]
Starting >>> yahboomcar_ctrl
Finished <<< rf2o_laser_odometry [3.69s]
Starting >>> yahboomcar_description
Finished <<< yahboomcar_base_node [3.78s]
Starting >>> ydlidar_ros2_driver
Finished <<< ydlidar_ros2_driver [3.71s]
Starting >>> slam_gmapping
Finished <<< slam_gmapping [1.33s]
Finished <<< yahboomcar_bringup [12.8s]
Finished <<< yahboomcar_description [12.6s]
Finished <<< yahboomcar_ctrl [12.8s]

Summary: 8 packages finished [19.3s]
sunrise@ubuntu:~/yahboomcar_ws$
```

4. Add feature package path to environment variables

Input following command in the terminal

```
sudo gedit ~/.bashrc
```

Paste the following content at the end of the file

```
source ~/yahboomcar_ws/install/setup.bash
```

5、 Save, exit, enter the following command to refresh environment variables

```
source ~/.bashrc
```

6、 Compiling dependencies that may need to be installed

```
#以ros-humble为例  
sudo apt install ros-humble-geographic-*  
sudo apt install libgeograph*
```